

Initial Project Planning Template

Date	03 July 2026
Team ID	N/A
Project Name	OptiCrop – Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Product backlog and sprint schedule for building OptiCrop.

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Data Collection & Preprocessing	USN-1	As a researcher, I can load and explore the crop dataset (N, P, K, temperature, humidity, pH, rainfall) so that I understand its structure before modeling.	2	High	—	01 Jun 2026	07 Jun 2026
Sprint-1		USN-2	As a researcher, I can visualize feature distributions (univariate/bivariate/multivariate) so that I can understand data patterns before modeling.	2	High	—	01 Jun 2026	07 Jun 2026
Sprint-1		USN-3	As a researcher, I can clean null values and handle outliers (e.g., in Potassium) so that the dataset is reliable for training.	2	Medium	—	01 Jun 2026	07 Jun 2026

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1		USN-4	As a developer, I can split the data into training and testing sets so that model performance can be evaluated fairly.	1	High	—	01 Jun 2026	07 Jun 2026
Sprint-2	Model Building	USN-5	As a developer, I can train multiple ML algorithms (KNN, Logistic Regression, Decision Tree, Random Forest) so that I can compare their performance.	3	High	—	08 Jun 2026	14 Jun 2026
Sprint-2		USN-6	As a developer, I can compare model performance using accuracy, precision, recall, and F1-score so that I can select the best model.	2	High	—	08 Jun 2026	14 Jun 2026
Sprint-2		USN-7	As a developer, I can save the best-performing model and scaler using Pickle so that predictions don't require retraining.	1	High	—	08 Jun 2026	14 Jun 2026
Sprint-2		USN-8	As a researcher, I can explore agricultural condition clusters using K-Means so that I can identify natural groupings in the data.	2	Low	—	08 Jun 2026	14 Jun 2026
Sprint-3	Application Building	USN-9	As a farmer, I can view the OptiCrop home and about pages so that I understand what the tool does.	1	Medium	—	15 Jun 2026	21 Jun 2026
Sprint-3		USN-10	As a farmer, I can enter soil and climate values in a form so that I can request a crop recommendation.	2	High	—	15 Jun 2026	21 Jun 2026
Sprint-3		USN-11	As a farmer, I can get an instant crop recommendation after submitting the form so that I can decide what to plant.	3	High	—	15 Jun 2026	21 Jun 2026

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-3		USN-12	As a farmer, I want the interface to be clean and easy to use on any device so that I can use it without technical difficulty.	2	Medium	—	15 Jun 2026	21 Jun 2026